

# 1. Waterfall and Scrum processes

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Chosen Product: Monobank

## 1.1. Waterfall

- **Requirements:**

### *Customer support*

An app must have a feature that allows the user to ask for help regarding how to use the app and other features. It should be 24/7 and must give an instant reply. For such features, AI is usually used, like intelligent chatbots. This would allow the user to have assistance anytime and anywhere.

### *Security*

The app can perform payments and transactions and is usually linked to the user's bank account. So the app must ensure secure sign-in and before the transaction, it must ask for some sort of password pin. It must have multi-factor authentication but it shouldn't mean that it becomes too time-consuming or complicated to do. It also must provide a method to regain the authority of the account in case the passwords have been forgotten.

### *Bank account management,*

After all, it's a mobile banking app and it primarily focuses on banking thus it must have a feature that allows users to monitor their cards and bank accounts, check their balances and transaction histories, and make transactions happen.

### *ATM Locator*

The app must have a feature that helps the user with locating the nearest ATM or a branch.

### *QR code scanning*

The app also must have a feature that would ensure fast payments. QR code scanning is usually famous and searched for the feature that allows the user to undergo an easy, fast and convenient way to perform the transaction. Usually, such apps have in-built QR code scanners.

### *Notifications*

Anyone who uses a smartphone knows how important notifications are and would agree that every app must have this feature. But it must also allow the user to stop receiving a notification if they wish so. The app should allow the user to choose what information they need to be notified for.

For the app to reach a level higher than a regular mobile banking app, it can add extra features such as...

- Tracking spending habits.
- Scheduled bills and payments.
- Cashback.
- Special offers.
- Shared finance features.
- Budgeting.

- **Design:** According to the best practices in banking app UI/UX design, we should deliver a personalized customer experience, make the account useful and helpful, strive for ease of use, and ensure the overall app is transparent and safe. At this stage, we should create specifications that describe how our product will work, choose a programming language, and choose where our data will be saved.
- **Implementation:** You can implement the app with an in-house team-chosen programming language. If you have the time and budget to hire banking software development professionals.
- **Verification:** We must check 5 central points: Bank Transfers, Tariffs, Blocking scenarios, User profile functionality, and Limit checking.

To approach the testing of banking applications and financial planners correctly, one should take into consideration the right sequence of the following steps:

1. **Requirements analysis.** As with any other testing, quality testing for banking applications should begin with collecting all the requirements. At this stage, it is important to take into account all details about the application, namely its goal and the full list of tasks it is designed to perform.
  2. **Database testing.** Since financial planners store a large amount of data, the QA engineer's responsibility is to check the maximum number of procedures linked with the database. More specifically, it is needed to check data integrity, loading, transfer, and checks of stored procedures, and functions.
  3. **Functional testing.** At this stage, the QA engineer prepares a test case, validates the test case, and executes it.
  4. **Security testing.** During preparation for testing, the QA team must include both negative and positive test scenarios to infiltrate the system and report it before any unauthorized person can access it.
  5. **Usability testing.** This type of testing is needed to ensure that customers with different abilities can use the system like a normal user.
  6. **User acceptance testing.** This is the final stage of end-user testing to ensure that the application matches the real-world scenario.
- **Maintenance:** Support for Monobank, ability to release new versions.

## 1.2. Scrum

Scrum increment 1: Create the customers' profiles

- **Requirements:** The customers' profiles have to be active personal information fields: name, gender, address, birth date, etc.; profile photo downloading/updating; navigation through app's sections with no Internet connection; editing data with no Internet connection.
- **Design:** Create a specification for this module, where we describe the architecture of the customers' profiles module, which language programming we will use, GUI, and how the module will work.
- **Implementation:** Create the module using language programming.
- **Verification:** testing the customers' profiles model. Checking requirements and the following functions: active personal information on fields: name, gender, address, birth date, etc.; profile photo downloading / updating; navigation through app's sections with no Internet connection; editing data with no Internet connection. Use module testing and integration testing.

Scrum increment N: Create the module of money transfers

- **Requirements:** The users will be able to: transfer money to a phone/card (of the same or another bank); request a transfer; return a transfer if it is not received yet; repeat the transfer sent before; make service/tariff payments; find the nearest ATM location.
- **Design:** Create a specification for this module, where we describe the architecture of the money transfers module, which language programming we will use, GUI, and how the module will work.

- **Implementation:** realize the module using a programming language.
- **Verification:** You should thoroughly examine all the money transfers on all the possible mobile devices and platforms. Check up how the next functions work: transfer money to phone/card (of the same or another bank), request a transfer, return a transfer if it is not received yet, repeat the transfer sent before, make service/tariff payments, find the nearest ATM location.

## 2. Methodologies pros and cons

| Methodology name | Pros                                                                                                                                                                                                                               | Cons                                                                                                                                                                                                                                                                                     |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agile            | <p>The project is divided into short iterations.</p> <p>Quickly get the first working version</p> <p>Minimize risk with a flexible change process</p> <p>Adjustment functional requirements built into the development process</p> | <p>It is impossible to calculate the exact amount of work due to constantly changing requirements</p> <p>The team must be highly professional and experienced</p> <p>New requirements may conflict with existing architecture</p> <p>There is a risk that the project will never end</p> |
| Waterfall        | <p>Clear process structure</p> <p>High-quality and detailed documentation</p>                                                                                                                                                      | <p>Requirements are defined and fixed at the very beginning, which deprives the process of flexibility</p>                                                                                                                                                                               |

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|  | <p>In the process, it is easy to track resources, risks and time spent</p> <p>Product tasks are stable from start to finish</p> | <p>The project takes a lot of time and resources.</p> <p>The client will see only the finished product, it is difficult to change</p> <p>Weak feedback between stages</p> |
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